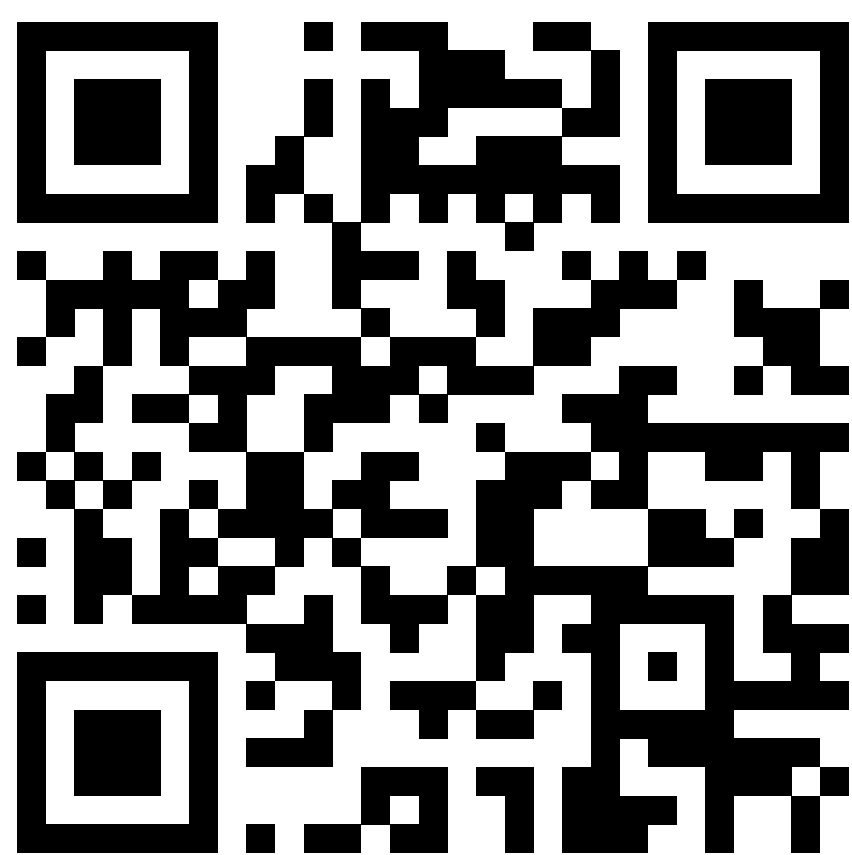


Early Childhood Development and Structural Change

Selim Elbadri

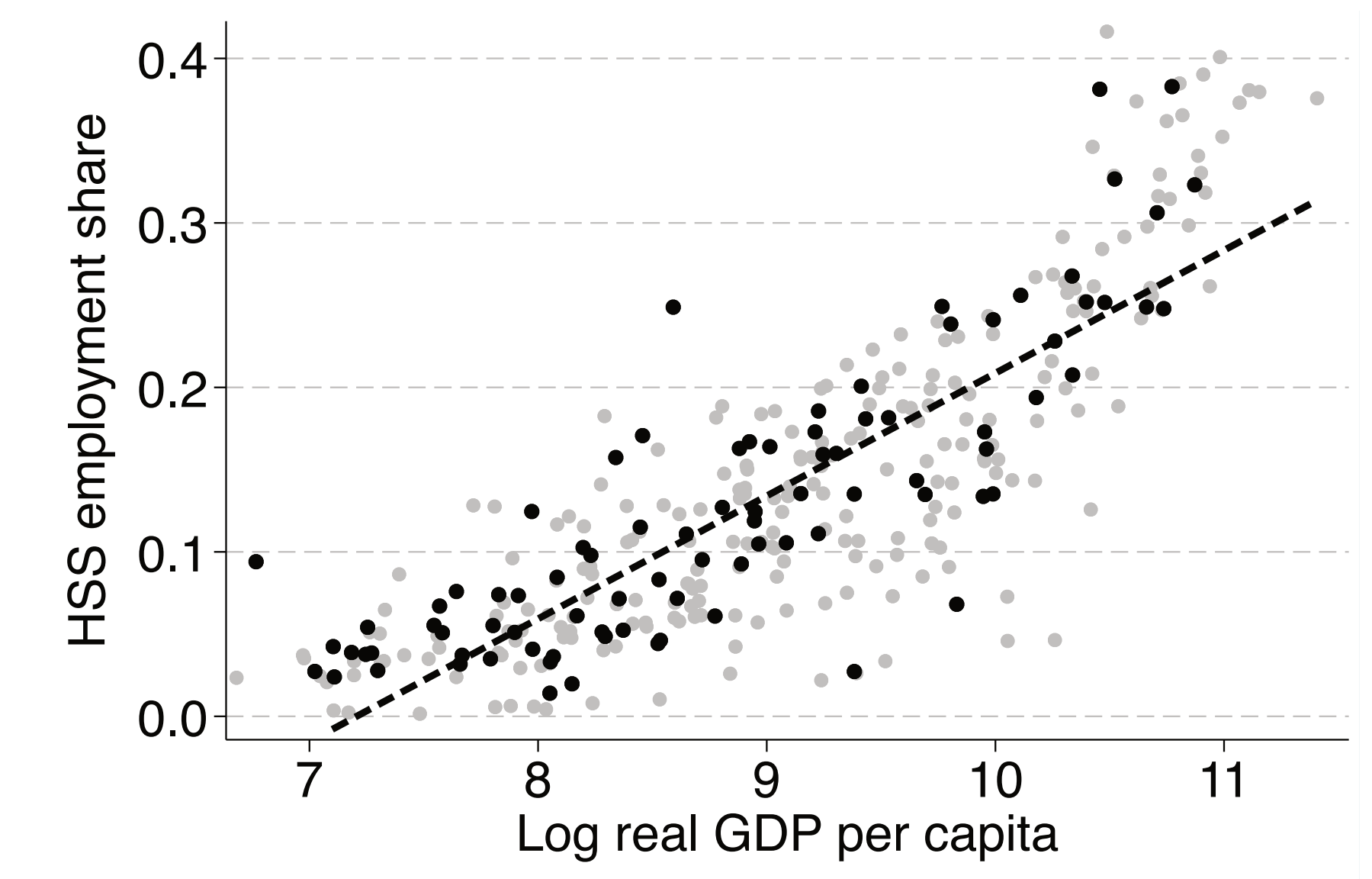
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Read the paper

The Research Question

Richer countries shift into **High-Skill Services** (HSS), the services that sustain high living standards.



Data: IPUMS International, PWT

This needs skills, whose supply is taken as given (Buera et al. 2022).

How does early childhood development (ECD) shape structural change?

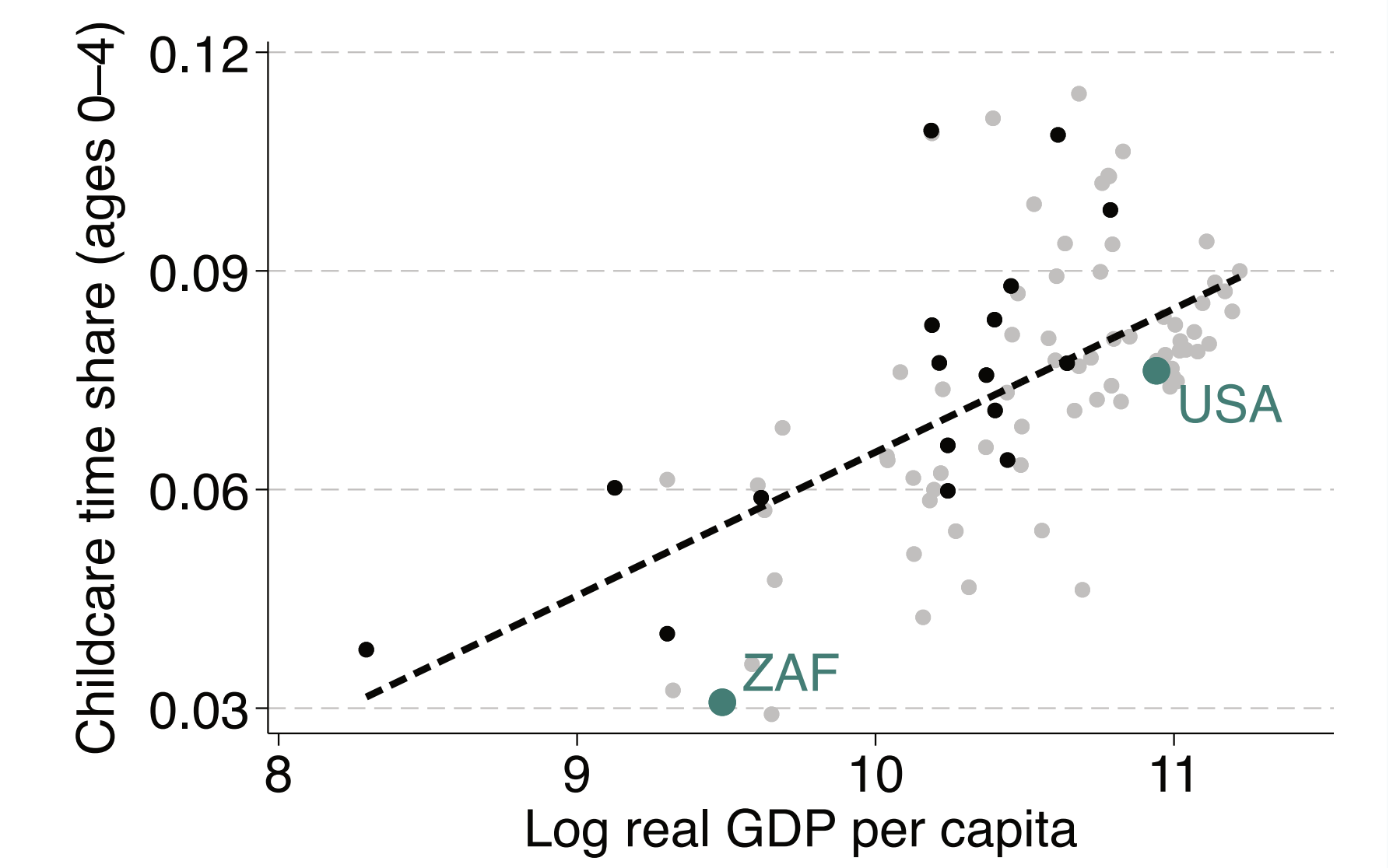
This paper: makes the supply of skills **endogenous**. Parents' time with young children rises with income, and early skills make later investment more productive, so development raises its own supply of skills.

Why it matters: skills built a generation ago limit how fast economies can move into HSS.

Time w/ Young Kids ↑ w/ Income

NEW FACT

Americans earn 4x as much as South Africans; they spend 2.5x as much time with young kids.

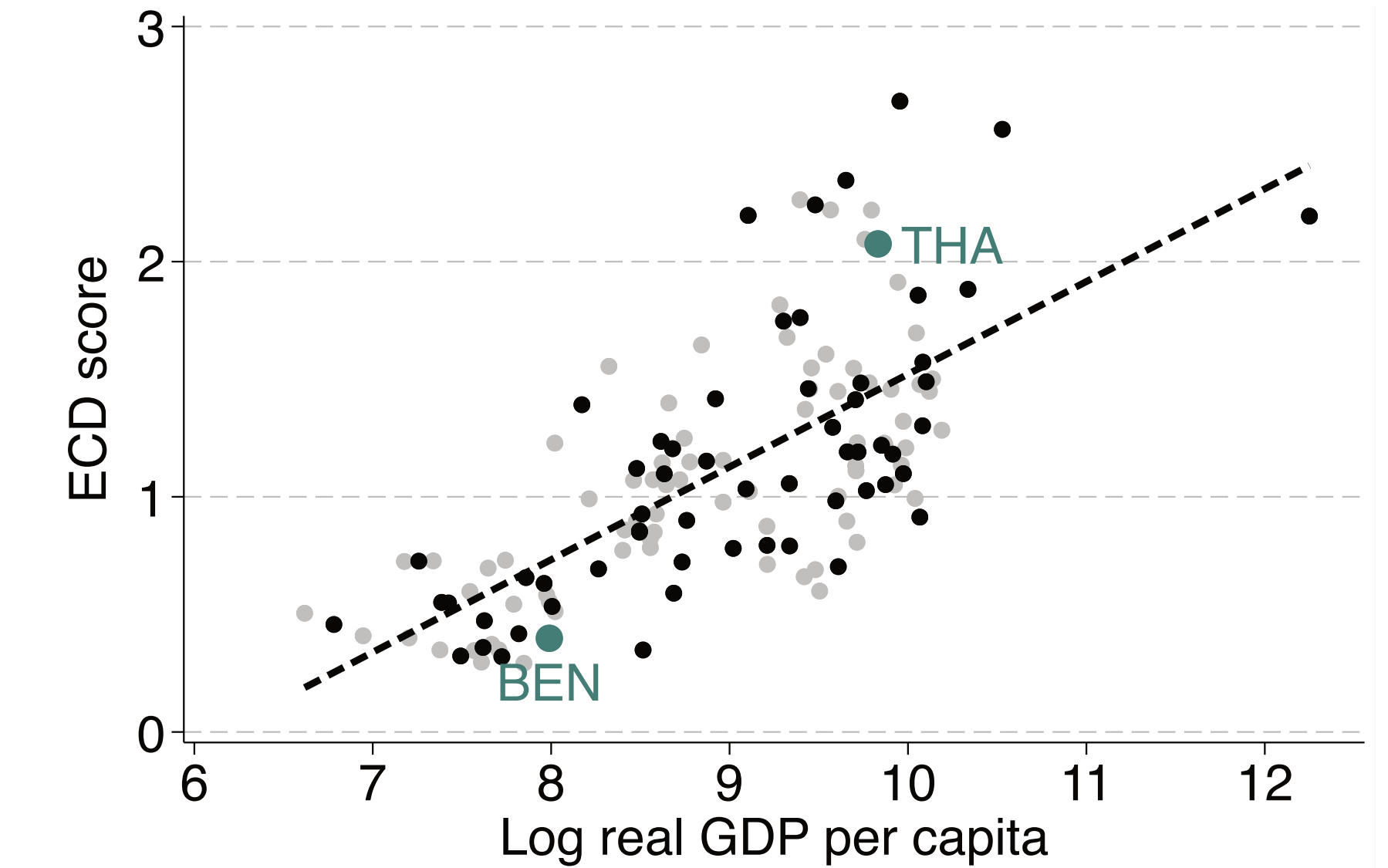


Data: Multinational Time Use Survey, PWT

ECD Outcomes ↑ w/ Income

NEW FACT

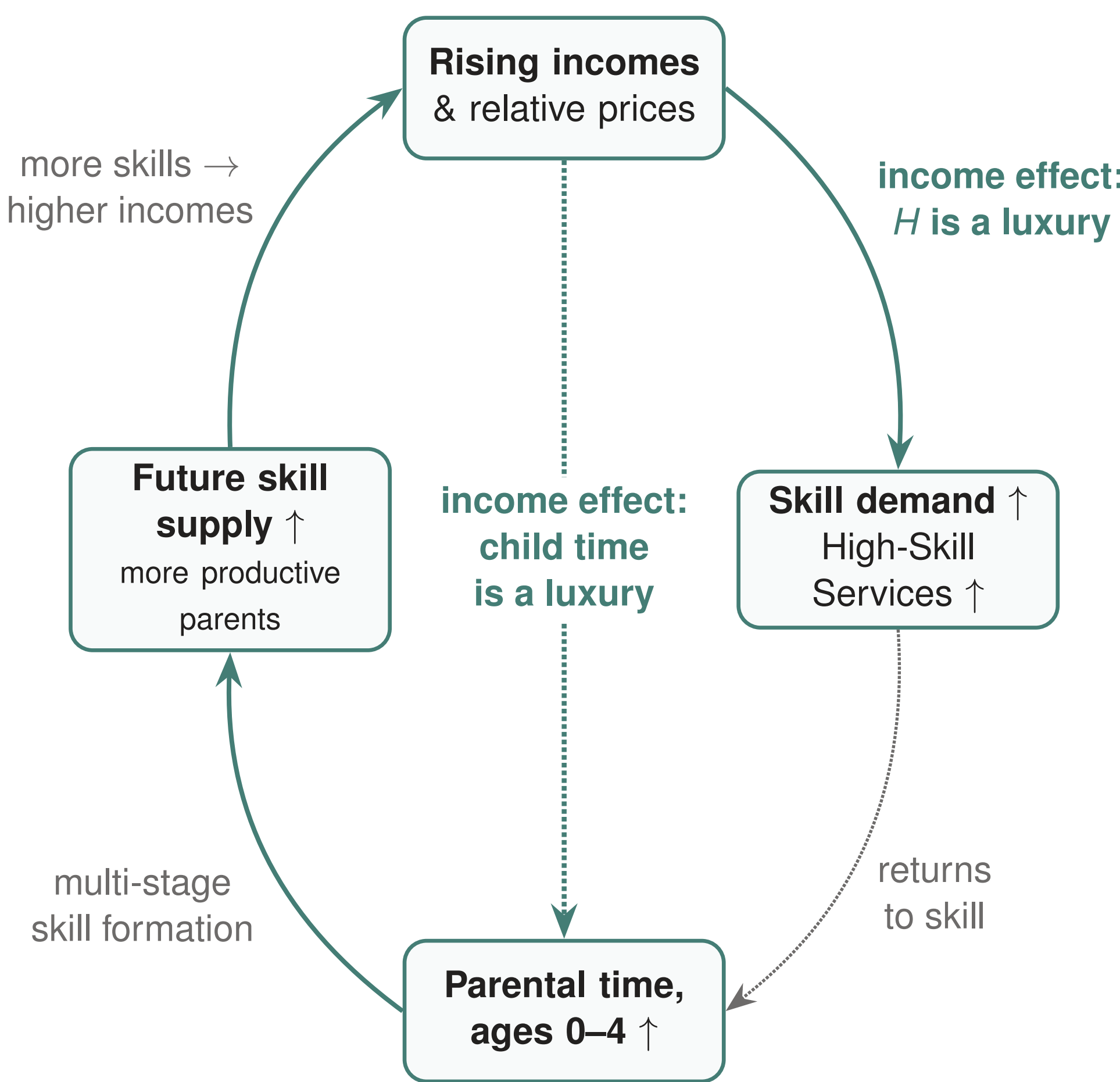
Thais earn 6x as much as Beninese; their 4-year-olds hit 5x as many milestones.



Data: IPUMS MICS, PWT

A Heterogeneous Agent GE Model

Heterogeneous dynasties and three sectors: Goods (G), Low-Skill Services (L) and High-Skill Services (H).



1. Preferences: Two Income Effects

Parents split time between work n and childcare τ , with $n = 1 - \tau$:

$$U = \frac{c^{1-\nu}}{1-\nu} - \omega \frac{n^{1+1/\chi}}{1+1/\chi}, \quad \sum_j \theta_j^{1/\sigma} \left(\frac{c_j}{c^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1.$$

With $\nu > 1$, childcare time rises with income. With $\lambda_H > \lambda_G$, demand shifts toward High-Skill Services.

2. Human Capital: Dynamic Complementarity

In each childhood stage $s \in \{1, \dots, S\}$, the child's skills h grow with the parent's effective time $h^p \tau$:

$$h_{t+1} - h_t = h_t^{\zeta_s} (h^p \tau_t)^{1-\zeta_s}.$$

Dynamic complementarity: existing skills raise the productivity of parental time investments.

3. Production: Skill Intensity and Uneven Growth

Each sector combines unskilled (N) and skilled (S) labour with a common CES:

$$Y_j = A_j \left[\phi_j N_j^{\frac{\epsilon-1}{\epsilon}} + (1-\phi_j) S_j^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \quad A_j(t) = A_j(0) e^{g_j t}.$$

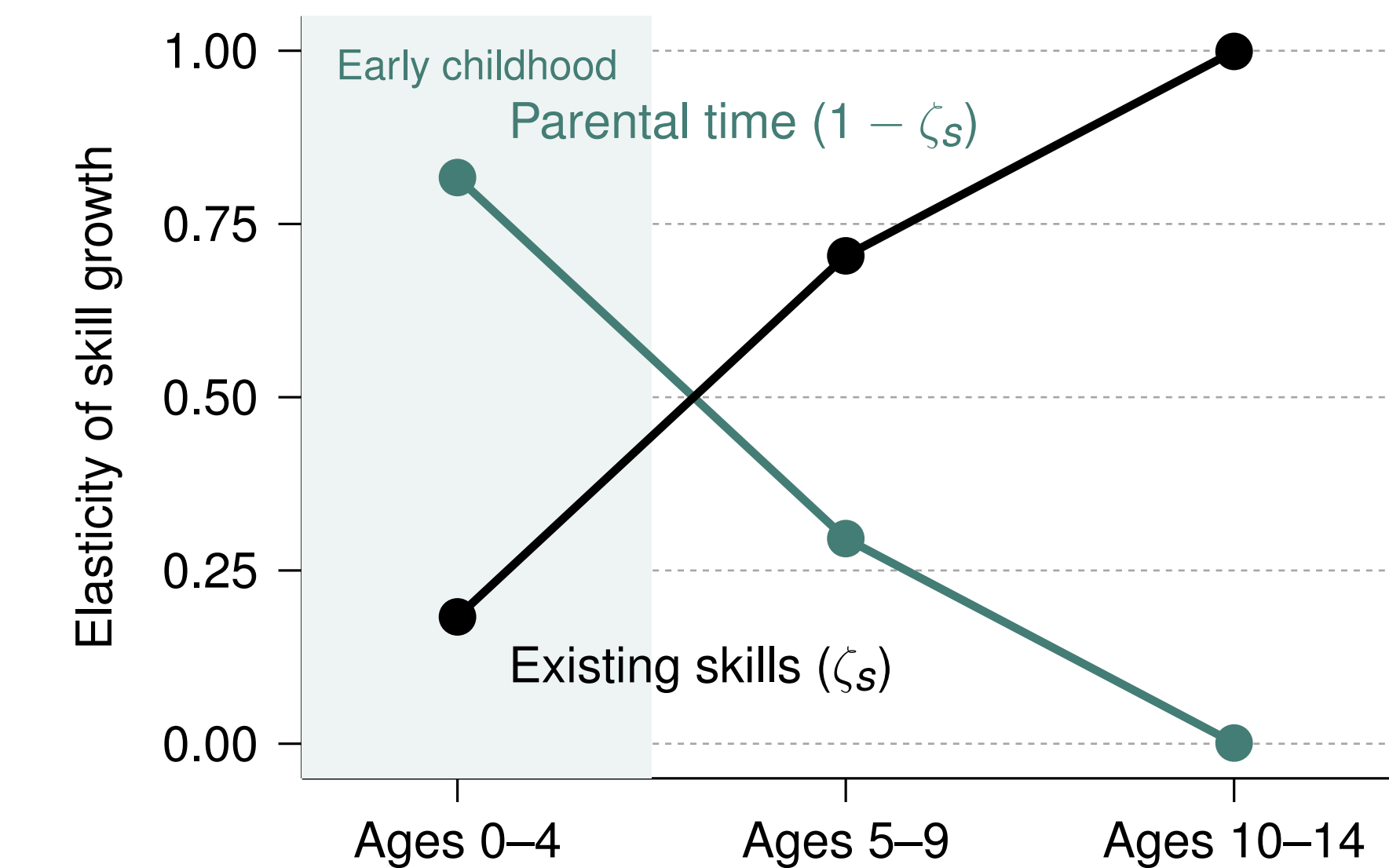
Sectors differ in skill intensity and TFP growth g_j :

$\phi_G = 1$	$0 < \phi_L < 1$	$\phi_H = 0$
Goods unskilled only	Low-Skill Services both	High-Skill Services skilled only

Calibration: Timing Matters

Matches postwar US structural change, time use and skill distribution.

Parental time matters most early; later skill growth builds on skills already formed.

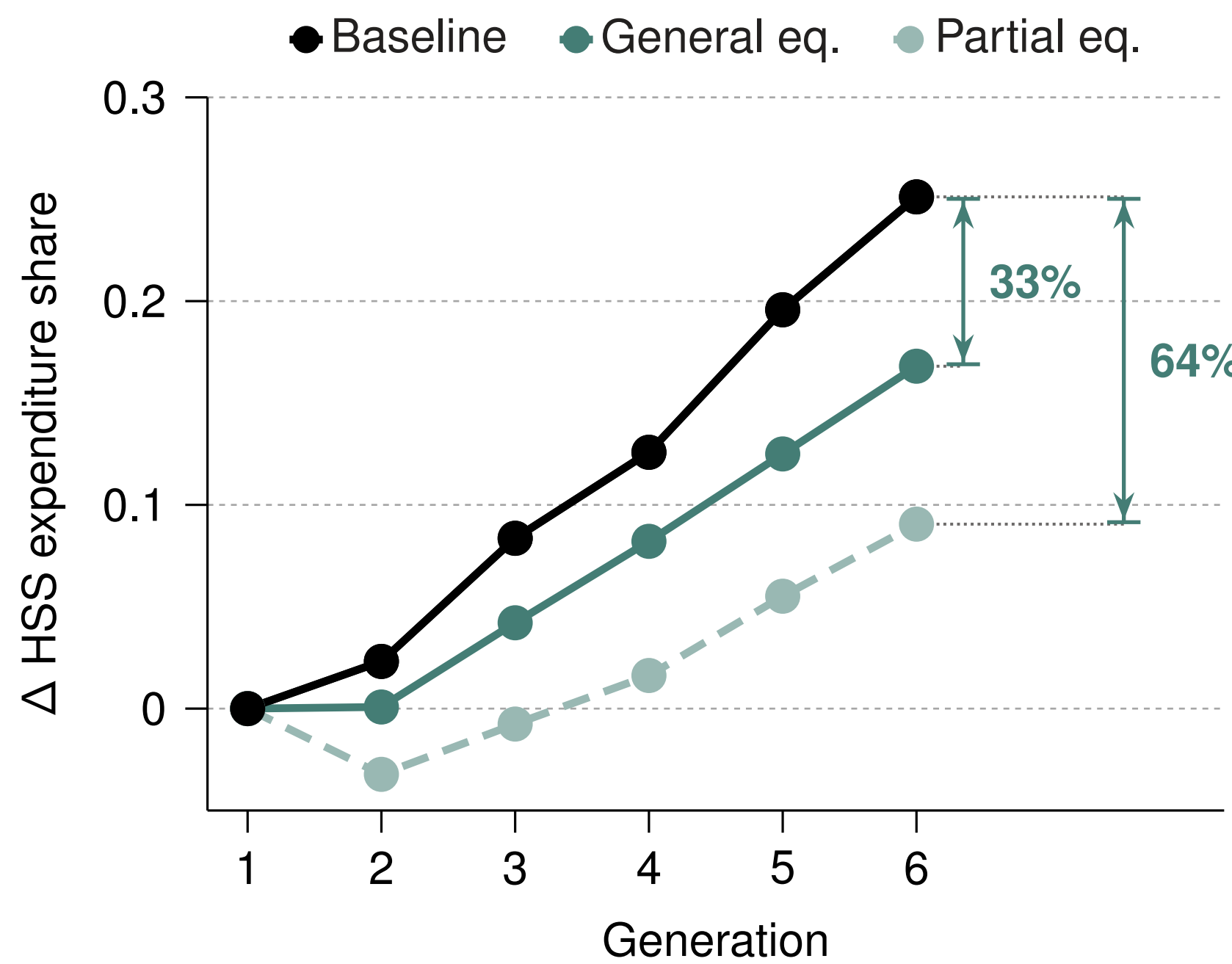


Source: Simulated Method of Moments

Dynamic Complementarity Matters

Experiment. Skill growth depends on the child's initial skills h_0 instead of current skills h_t , so earlier investment no longer compounds.

The US loses a third of its shift into High-Skill Services, and nearly two-thirds with prices and wages fixed.



% = share of the baseline rise (25.1 p.p.) lost

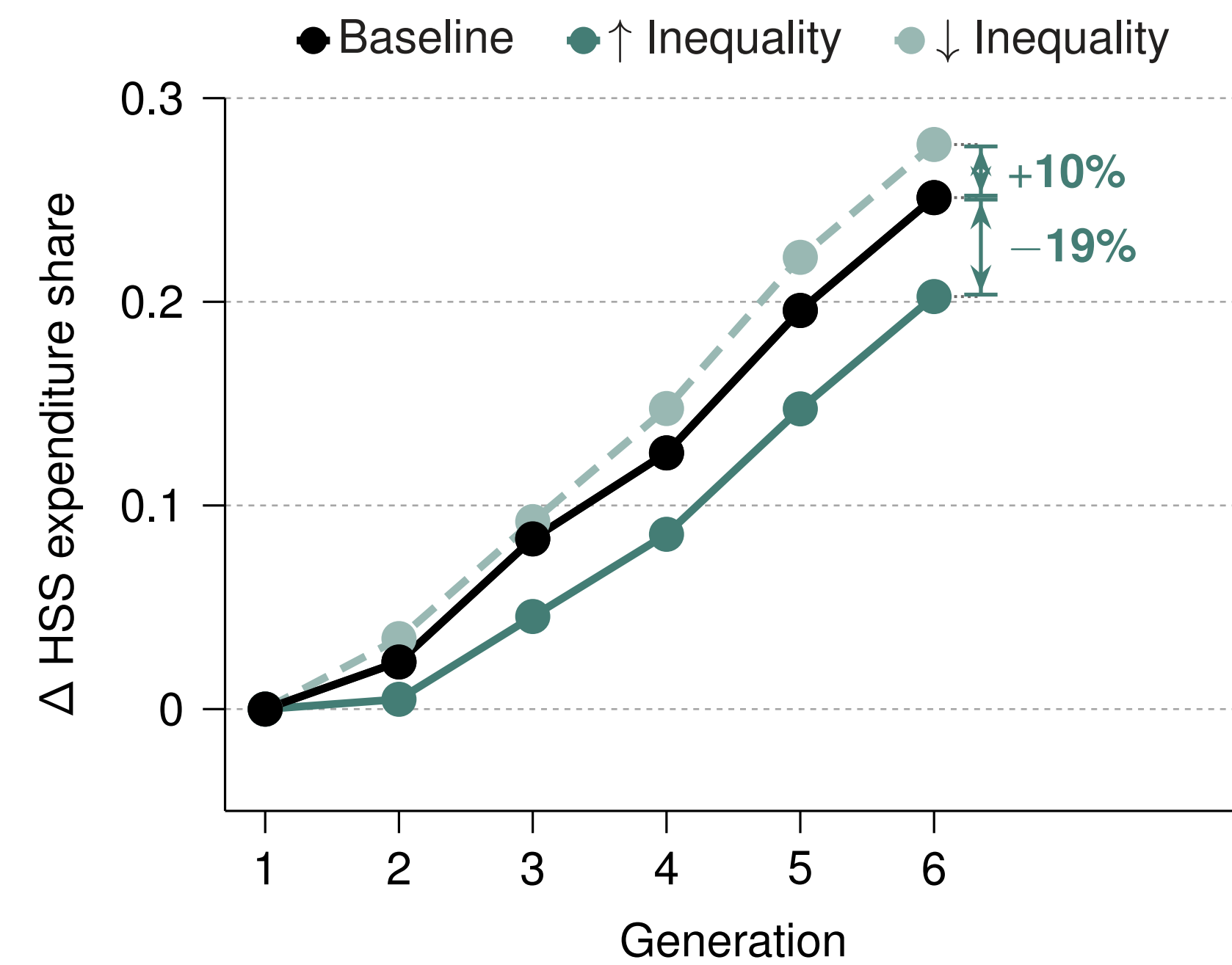
Why? Skilled workers become less skilled

- Early investment no longer compounds, so few children become skilled.
- In GE, a higher skill premium pulls less-skilled workers into skilled jobs: their **number** recovers, but their **human capital** falls from 4.21 to 0.69.

Inequality Slows the Shift

Experiment. Mean-preserving spread of initial human capital (children's draws and first-generation parents), making it 35% more (or less) dispersed.

More inequality cuts the shift into High-Skill Services by a fifth; less inequality speeds it up by a tenth.



% = change in the baseline rise (25.1 p.p.)

Why? Again, quality

- Parents invest less time in young children across the distribution.
- The **number** of skilled workers barely moves, but their **human capital** falls from 4.21 to 1.92.

Takeaway

1/3 of the US shift toward HSS runs through ECD via dynamic complementarity.